Simulation Section Draft

V. Numerical Results

We evaluate the framework through three complementary experiments. First, image-classification comparisons examine deployment accuracy and bounded risk under analog fading and additive noise. Second, controlled feature-erasure experiments isolate the effect of a finite training channel bank and the choice of training channel law. Finally, we evaluate the channel-shift, channel-complexity, and data-complexity contributions at frozen learned posteriors and compute a numerical population-risk bound. ERM and L2 are the main competing baselines. Supplementary results record additional internal ablations and sensitivity checks.

A. Experimental Setup

1) Data and split networks: The image experiments use MNIST [1] and Fashion-MNIST [2]. Both contain ten classes of 28×28 grayscale images. For MNIST, we use 55,000 training and 5,000 validation examples from the official training partition. For Fashion-MNIST, we use a subset of 10,000 training examples and a disjoint validation set of 5,000 examples. Both use the official 10,000-example test partition. Pixel intensities are mapped to $[0, 1]$, without augmentation or fitted normalization. The common image classifier has dimensions $784 \to 128 \to 10$. Its first affine layer and ReLU form the transmitter; its final affine layer forms the receiver, with transmission between the two.

We additionally evaluate CIFAR-10 [3], which contains ten classes of 32×32 RGB images. Its official training partition is split into 45,000 training and 5,000 validation examples; the official test partition contains 10,000 examples. Pixels are mapped to $[-1, 1]$ by the fixed transform $x/127.5 - 1$. The transmitter uses three 3×3 convolution/ReLU blocks with 16, 32 and 64 output channels, max pooling after the first two blocks, adaptive 4×4 pooling, and an affine/ReLU projection to 128 transmitted features. A ten-class affine receiver follows the channel. All methods share this 156,074-weight architecture, without batch normalization, dropout or augmentation. This extends both the dataset and transmitter architecture; it is not an isolated architecture ablation.

For controlled experiments, we sample an i.i.d. noisy two-moons population: the binary label is Bernoulli with parameter $1/2$, the arc angle is uniform on $[0, \pi]$, and independent Gaussian coordinate noise has standard deviation 0.3. These experiments use $2 \to d \to 2$ split networks. The finite-channel study fixes $d = 8$, $n = 512$, a 512-example validation set, and an independent 10,000-example test set. The training-law study uses $d = 16$, 800 training examples, and 200 validation examples. Table I summarizes the main settings.

2) Channel laws and finite observations: For analog transmission, the recovered representation is $\widehat{f} = Hf + N$, where H is diagonal with independent $H_{kk} \sim \mathcal{N}(\mu_m, \sigma_m^2)$ and $N_k \sim \mathcal{N}(0, \sigma_b^2)$. The image comparisons use $(\mu_m, \sigma_m, \sigma_b) = (0.5, 1, 1)$. These are fixed distortion-law parameters: feature power is not normalized, so the comparisons are not at a common physical SNR or transmit-power constraint. Neither CSI nor equalization is supplied to the receiver.

For feature erasures, $\widehat{f}_k = B_k f_k$, where the coordinates $B_k \sim \text{Bernoulli}(1 - p)$ are independent. The receiver observes zero-filled features without a separate erasure-indicator input. This is an erasure abstraction for continuous features, rather than a complete bit-level digital transmission system. A channel-bank entry is one full gain/noise vector or one full erasure mask. The bank is sampled separately from the data and reused throughout training. Each training example in a minibatch receives an independent uniform bank index, giving an unbiased stochastic estimate of the empirical risk on $S \times U$.

3) Bounded loss and optimization: Let $p_w(y \mid x, c)$ be the confidence assigned to the correct class. With $\varepsilon_\ell = 0.01$, we use the common smooth bounded logarithmic loss

$$\ell_{\varepsilon_\ell}(w, c, z) = \frac{-\log[\varepsilon_\ell + (1 - \varepsilon_\ell) p_w(y \mid x, c)]}{\log(1/\varepsilon_\ell)}. \tag{1}$$

It lies in $[0, 1]$ uniformly over weights, channels, and data. Hence the standing sub-Gaussian assumptions hold with $\sigma^2 = \sigma_{\text{tr}}^2 = 1/4$ by Hoeffding's lemma. For perfect training, the sharper channel variance proxy is zero.

The proposed learner uses one diagonal Gaussian posterior and the fixed, data-independent prior $Q_W = \mathcal{N}(0, I)$. Its weight means start from fan-in-scaled uniform initialization with zero biases; posterior standard deviations initialize at 0.003 or 0.01 in the grayscale experiments, and at 0.001 or 0.003 in the CIFAR-10 search. We optimize with Adam and a constant learning rate, drawing one weight sample per minibatch. Grayscale-image batches contain 256 examples; CIFAR-10 and toy batches contain 128. The T1 objective uses the separate penalties in Theorem 1; T2 uses the joint penalty in Theorem 2. Unless a joint confidence allocation is specified below, the nominal failure budget is 0.05, split equally between T1's two terms. ERM and L2 use the same loss (1); L2 adds an explicit squared-weight penalty.

4) Selection and evaluation: The explored configurations are selected by mean validation accuracy over the listed initialization/channel-bank seeds, breaking ties by lower cross entropy. Checkpoint selection uses two fixed validation random-number streams across epochs; test evaluation uses fresh streams. These are exploratory numerical comparisons, and all listed seeds are retained in the main tables. The data split is fixed across these seeds. Thus the reported standard deviations measure initialization/channel-bank variation conditional on that split. Search ranges and selected settings are provided in the supplement.

Test accuracy is the Gibbs-expected accuracy: a weight draw is held fixed for one complete test pass, with a fresh channel for each example, and the resulting accuracy is averaged over ten passes. ERM and L2 are deterministic, so

TABLE I
SETTINGS OF THE MAIN EXPERIMENTS

Experiment	n	Validation	Test	d	m	Epochs	Seeds
MNIST, analog	55,000	5,000	10,000	128	100	100	3
Fashion-MNIST, analog	10,000	5,000	10,000	128	100	25	2
CIFAR-10, analog	45,000	5,000	10,000	128	100	100	3
Two moons, finite bank	512	512	10,000	8	5–2000	150	3
Two moons, training-law mismatch	800	200	10,000	16	100	150	3

only their channels vary. Bounded loss is evaluated using the same posterior convention; ordinary cross entropy is a separate metric. Perfect training means identity transmission in the training forward pass, followed by testing through the deployment channel. For Proposed Perfect, the objective retains only the data PAC-Bayes penalty. Checkpoint selection for these references uses deployment-channel validation; the perfect-training label refers to the training forward channel.

B. Deployment Performance Under Analog Fading

Table II reports mean test accuracy and normalized bounded loss. On MNIST, T1 attains 76.63%, compared with 74.60% for matched ERM and 70.12% for matched L2, giving gains of 2.03 and 6.52 percentage points. T2 reaches 76.43% and slightly lowers the bounded loss relative to T1. On Fashion-MNIST, T1 and T2 attain 63.24% and 63.21%, respectively, versus 61.57% for ERM and 61.49% for L2. Both proposed variants outperform these evaluated baselines in every reported seed of the two image comparisons. Their small differences in accuracy and bounded loss do not imply a universal ordering between T1 and T2.

The perfect-training references also expose the importance of training with channel observations. On MNIST, Proposed Perfect attains 57.94% under deployment, whereas matched T1 attains 76.63%; the corresponding Fashion-MNIST accuracies are 52.44% and 63.24%. Removing the finite-channel penalty during perfect training therefore does not remove the deployment mismatch. These results concern the tested configurations; they do not assert optimality over all baseline hyperparameters.

1) CIFAR-10 with a convolutional transmitter: Table III extends the comparison to the shared convolutional architecture. The 34-run preliminary screen selects one setting per method by aggregate validation accuracy. We then train these seven fixed settings for 100 epochs on 45,000 examples using three new initialization/channel-bank seeds; all three are retained. T1 achieves 55.03% mean test accuracy, versus 53.67% for L2 and 46.16% for ERM, gains of 1.36 and 8.87 percentage points. It exceeds both baselines in every reported seed and also has the lowest mean bounded risk, 0.3395, compared with 0.3526 for L2 and 0.4205 for ERM. T2 attains 53.26%, slightly below L2; the benefit therefore does not extend uniformly to both proposed variants. Mean transmitted feature power is 22.73 per coordinate for T1, versus 12.59 for L2 and 6.71 for ERM. These remain fixed-channel-law comparisons with unconstrained feature power.

Perfect-trained ERM, L2 and Proposed attain approximately 34% under the same deployment law, reinforcing the training/deployment distinction. The full-data comparison uses a fixed 100-epoch budget; several selected checkpoints occur near its end, so these results do not establish fully converged or globally optimal baseline accuracy. The preliminary screen and search settings are recorded in the supplement. The high-dimensional CNN bound at $m = 100$ is vacuous; the numerical certificate below is computed separately on the controlled small network.

C. Generalization From a Finite Channel Bank

For a frozen posterior, we evaluate bounded risk on the reused bank, $\widehat{R}_{S,U}$, on the same data with fresh training-law channels, $\widehat{R}_{S,\mathrm{tr}}$, and on held-out data with fresh training-law channels, $\widehat{R}_{\mathrm{te,tr}}$. The signed channel gap is

$$\widehat{G}_{\mathrm{ch}} = \widehat{R}_{S,\mathrm{tr}} - \widehat{R}_{S,U}. \tag{2}$$

This evaluates unseen channels without simultaneously changing the data.

Figure 1 varies $m \in \{5, 20, 100, 500, 2000\}$ while keeping the training budget at 600 minibatch updates. For T1, the observed channel gap decreases from approximately 0.0103 at $m = 5$ to 0.0016 at $m = 20$, and is close to zero from $m = 100$ onward. This illustrates the additional generalization effect induced by a finite channel bank even when the training and deployment distributions match. The proposed variants nearly overlap in this controlled example; the experiment supports the finite-channel mechanism rather than a strict ranking between their optimizers. Residual fluctuations at larger m are comparable to the finite-evaluation scale.

D. Choice of the Training Channel Distribution

Figure 2 fixes deployment and considers perfect training together with training erasure probabilities 0.1, 0.3, 0.5, 0.7, and 0.9. T1 achieves mean accuracies of 83.13%, 84.42%, 84.90%, 84.84%, and 84.27% at the five nonzero probabilities. The matched point also minimizes cross entropy in this experiment. Proposed Perfect achieves 81.87%, about three percentage points below matched T1. Thus a perfect training forward channel does not provide the best deployment accuracy in this setting. Mildly and severely mismatched training laws yield different results even though the deployment law remains unchanged. The theory guarantees that matching removes the shift penalty; it does not require matched training to maximize accuracy for every model and finite dataset.

TABLE II

DEPLOYMENT PERFORMANCE UNDER MATCHED GAUSSIAN FADING AND ADDITIVE NOISE. ACCURACY IS IN PERCENT; BOUNDED RISK USES THE SMOOTH LOSS IN (1). ENTRIES ARE MEAN ± SAMPLE STANDARD DEVIATION OVER THREE INITIALIZATION/CHANNEL-BANK SEEDS FOR MNIST AND TWO FOR FASHIONMNIST, WITH TEN FRESH GIBBS/CHANNEL DRAWS PER CHECKPOINT. PERFECT REFERS TO TRAINING ONLY; ALL MODELS FACE THE SAME DEPLOYMENT LAW. BOLD IDENTIFIES THE BEST REPORTED MEAN WITHIN EACH METRIC.

	MNIST ($n = 55{,}000$, $m = 100$)		FashionMNIST ($n = 10{,}000$, $m = 100$)	
Training method	Accuracy (%)	Bounded risk	Accuracy (%)	Bounded risk
ERM (perfect)	58.07 ± 0.42	0.2602 ± 0.0030	50.95 ± 1.30	0.3012 ± 0.0134
ERM (matched)	74.60 ± 1.12	0.1944 ± 0.0098	61.57 ± 0.09	0.2287 ± 0.0001
ℓ_2 (matched)	70.12 ± 1.21	0.1950 ± 0.0070	61.49 ± 0.42	0.2295 ± 0.0024
Proposed (perfect)	57.94 ± 0.60	0.2601 ± 0.0029	52.44 ± 0.29	0.2881 ± 0.0060
Proposed T1 (matched)	$\mathbf{76.63 \pm 0.44}$	0.1795 ± 0.0034	$\mathbf{63.24 \pm 0.21}$	0.2188 ± 0.0003
Proposed T2 (matched)	76.43 ± 0.76	$\mathbf{0.1792 \pm 0.0064}$	63.21 ± 0.17	$\mathbf{0.2187 \pm 0.0004}$

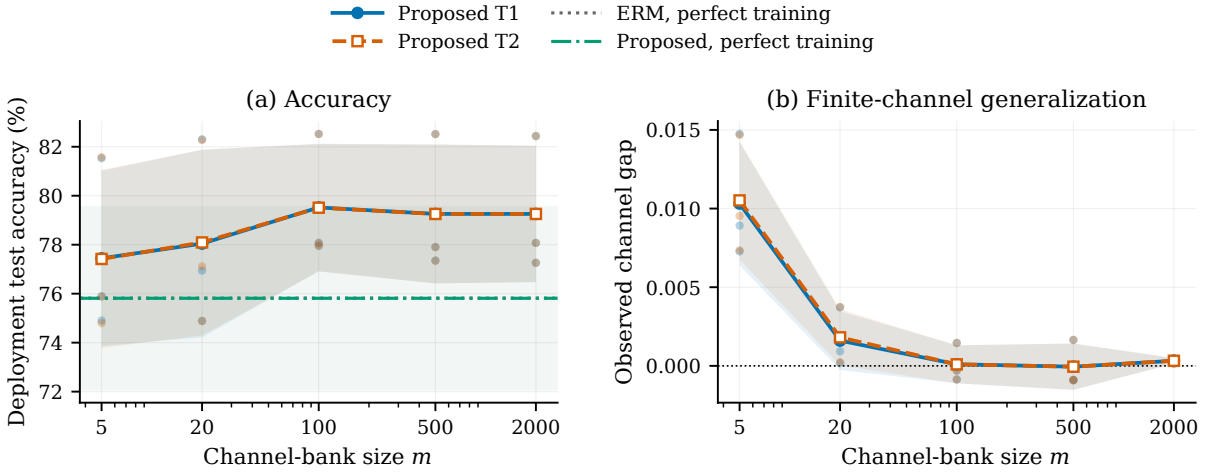


Fig. 1. Effect of the reusable channel-bank size under matched feature erasures: $p_{\mathrm{tr}} = p_{\mathrm{ch}} = 0.5$, $n = 512$, and $d = 8$. Curves show means over three initialization/channel-bank seeds; shading shows one sample standard deviation and points retain individual seeds. The data, number of optimizer updates, and deployment law are fixed across m. Perfect-trained references are reused across bank sizes. The right panel isolates the observed channel generalization gap in normalized-loss units.

TABLE III

CIFAR-10 DEPLOYMENT PERFORMANCE UNDER GAUSSIAN FADING AND ADDITIVE NOISE. ENTRIES ARE MEAN ± SAMPLE STANDARD DEVIATION ACROSS INITIALIZATION/CHANNEL-BANK SEEDS CONDITIONAL ON THE FIXED DATA SPLIT, USING TEN FRESH GIBBS/CHANNEL DRAWS PER CHECKPOINT. FULL DATA: $n = 45{,}000$, 100 EPOCHS, 3 PAIRED SEEDS. CONFIGURATIONS USE AGGREGATE VALIDATION ACCURACY; CHECKPOINTS USE VALIDATION ACCURACY. PERFECT REFERS TO THE TRAINING FORWARD CHANNEL ONLY. BOLD MARKS THE BEST MEAN WITHIN EACH METRIC.

Method	Full data	
	Accuracy (%)	Bounded risk
ERM (matched)	46.16 ± 1.33	0.4205 ± 0.0100
L2 (matched)	53.67 ± 0.73	0.3526 ± 0.0055
Proposed T1 (matched)	$\mathbf{55.03 \pm 0.29}$	$\mathbf{0.3395 \pm 0.0020}$
Proposed T2 (matched)	53.26 ± 0.60	0.3541 ± 0.0023
ERM (perfect)	34.44 ± 0.97	0.4438 ± 0.0086
L2 (perfect)	34.43 ± 0.92	0.4400 ± 0.0029
Proposed (perfect)	34.02 ± 0.89	0.4501 ± 0.0214

E. Bound Decomposition and Numerical Risk Guarantees

The held-out deployment estimate is denoted by $\widehat{R}_{\mathrm{te,ch}}$. We decompose its difference from the reused-bank empirical estimate into

$$\begin{aligned} \widehat{G}_{\mathrm{data}} &= \widehat{R}_{\mathrm{te,tr}} - \widehat{R}_{S,\mathrm{tr}}, \\ \widehat{G}_{\mathrm{shift}} &= \widehat{R}_{\mathrm{te,ch}} - \widehat{R}_{\mathrm{te,tr}}, \\ \widehat{R}_{\mathrm{te,ch}} - \widehat{R}_{S,U} &= \widehat{G}_{\mathrm{ch}} + \widehat{G}_{\mathrm{data}} + \widehat{G}_{\mathrm{shift}}. \end{aligned} \tag{3}$$

Every term uses the same frozen posterior and paired weight draws. These signed estimates are distinct from the nonnegative upper-bound terms.

For the bounded loss, channel shift also satisfies $|\mathbb{E}_P g - \mathbb{E}_Q g| \le \mathrm{TV}(P, Q)$ for $0 \le g \le 1$. For independent feature erasures, total variation is evaluated exactly by grouping masks by their number of erased coordinates. Perfect training therefore gives the finite shift penalty $1-(1-p_{\mathrm{ch}})^d$, despite infinite channel KL when $p_{\mathrm{ch}} > 0$. This bounded-loss specialization does not require an estimated channel-Lipschitz constant.

Figure 3 separates the mechanisms. Perfect training has no channel complexity but retains deployment mismatch. Matched training removes the shift penalty while retaining finite-channel and data complexity. A stochastic mismatched training law generally retains all three terms. The signed measured gaps need not be positive. The displayed gap bounds exceed one and are vacuous on the normalized-loss scale; their purpose here is to distinguish the three contributions.

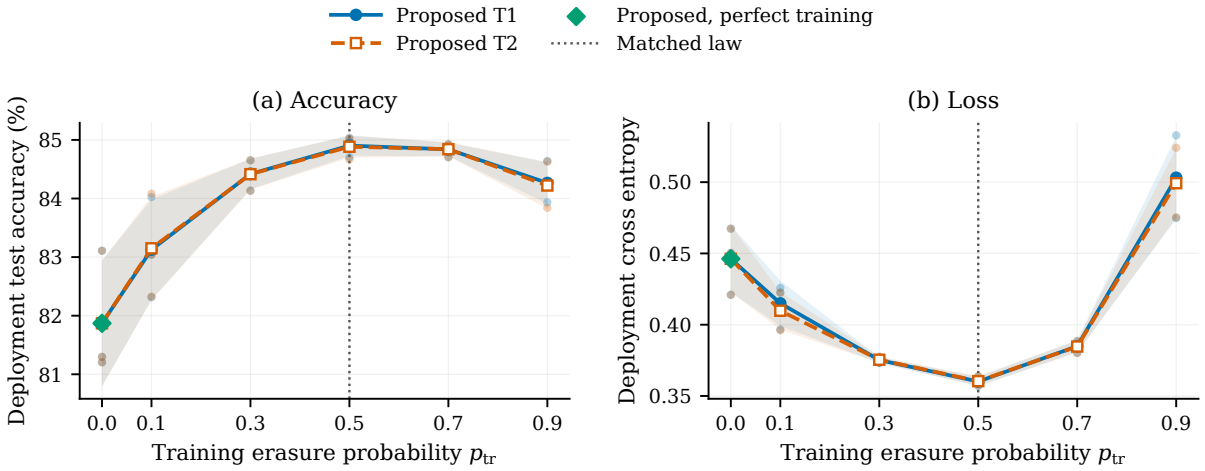


Fig. 2. Fixed deployment feature-erasure probability $p_{ch} = 0.5$ and varying training probability, with $n = 800$, $d = 16$, and $m = 100$. Perfect training is the point $p_{tr} = 0$; its data-only PAC objective is common to the proposed perfect-channel specialization. Means and one sample standard deviation use all three listed seeds. The vertical line marks matched training.

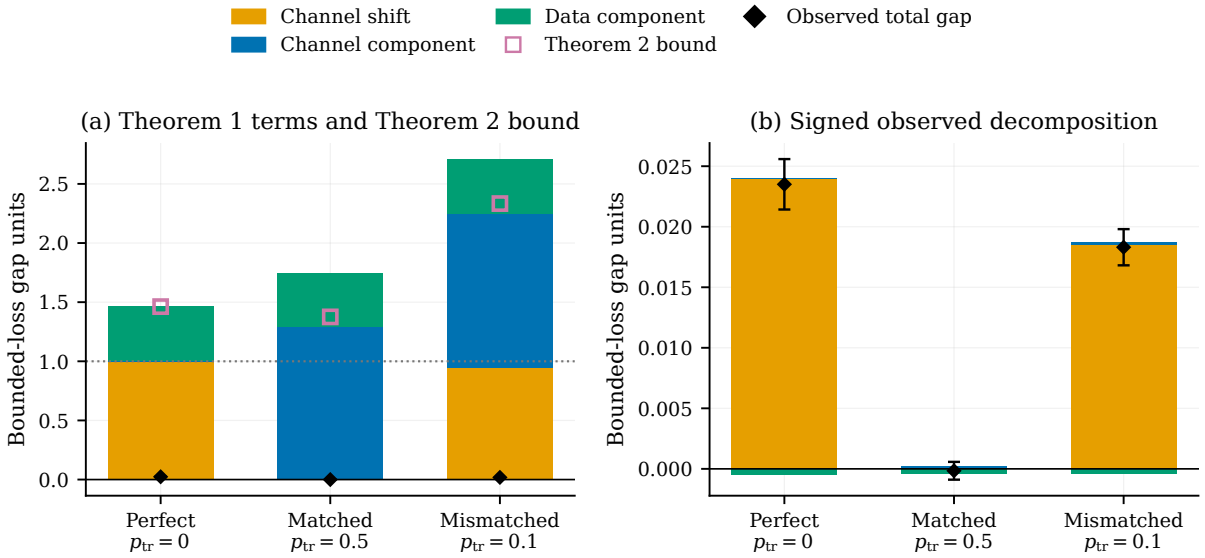


Fig. 3. Unweighted bound terms and signed observed gaps for perfect, matched, and mismatched training, at the same deployment law $p_{ch} = 0.5$, $n = 800$, $d = 16$, and $m = 100$. Each column evaluates the frozen posterior learned under its stated training law. Stacked theoretical terms are those of T1; markers evaluate T2 at the same posterior. The per-run nominal failure budget is 0.05. Shift uses the bounded-loss total-variation specialization. These illustrative gap estimates do not include the additional MC correction used in Table IV.

TABLE IV

TWO POPULATION-RISK BOUNDS EVALUATED AT THE SAME FROZEN T1-TRAINED POSTERIOR ($d = 8$, $n = 512$, $m = 2000$). A FRESH 100-DRAW EMPIRICAL-RISK ESTIMATE IS SHARED BY BOTH BOUNDS. THE TOTAL FAILURE BUDGET IS 0.05: 0.04 IS SPLIT ACROSS BOTH THEOREMS AND 21 DECLARED SETUPS, AND 0.01 CONTROLS MONTE CARLO ERROR. VALUES CONCERN NORMALIZED BOUNDED LOSS, NOT CLASSIFICATION ERROR.

Component	Theorem 1	Theorem 2
Empirical bounded risk estimate	0.090943	0.090943
Monte Carlo upper correction	0.151743	0.151743
Channel shift	0.000000	0.000000
Channel complexity	0.212937	–
Data complexity	0.420369	–
Joint complexity	–	0.475078
Population-risk upper bound	0.875991	0.717764

We additionally compute a corrected population-risk bound for a chosen T1-trained posterior with $n = 512$, $m = 2000$, and $d = 8$, using the same posterior for both theorem evaluations. A fresh 100-draw empirical-risk estimate is upper-bounded by Hoeffding's inequality with failure budget 0.01. To cover selection among the seven declared matched-erasure (n, m) settings and three bank seeds, we assign $0.04/(2 \times 21)$ to each theorem/setup event. The union of these events and the single fresh MC event has failure probability at most 0.05.

The corrected population-risk upper bounds are 0.8760 for T1 and 0.7178 for T2, including MC error. Both are below one. The joint bound is tighter at the same posterior; it does not change the predictor or imply an accuracy ordering. These bounds concern the normalized Gibbs loss in (1), rather than classification error or unbounded CE.

APPENDIX A
SUPPLEMENTARY NUMERICAL RESULTS

A. Additional Channel-Bank Comparisons

The reduced-data MNIST comparisons use 10,000 training examples, a hidden width of 128, and smaller analog-channel banks. They provide a sensitivity check rather than an independent test dataset: the official test set is the same as in the full-data comparison. Table V reports all seeds of the corresponding ten-draw cohorts. A separate five-draw screening cohort is retained in Table VI; its numbers are not pooled with the ten-draw evaluations.

TABLE V
REDUCED-DATA MNIST UNDER THE SAME ANALOG DEPLOYMENT LAW ($n = 10{,}000$, $d = 128$, 35 EPOCHS). ENTRIES USE ALL TWO PAIRED INITIALIZATION/CHANNEL-BANK SEEDS AND TEN FRESH DRAWS PER CHECKPOINT; SETTINGS WERE SELECTED ON VALIDATION DATA.

Condition	Method	Accuracy (%)	Bounded risk
$m = 20$	ERM	58.22 ± 0.82	0.3037 ± 0.0179
$m = 20$	ℓ_2	57.04 ± 0.92	0.3117 ± 0.0151
$m = 20$	Proposed T1	$\mathbf{59.89 \pm 0.20}$	$\mathbf{0.2886 \pm 0.0095}$
$m = 20$	Proposed T2	59.68 ± 0.06	0.2947 ± 0.0079
$m = 50$	ERM	66.04 ± 0.56	0.2427 ± 0.0046
$m = 50$	ℓ_2	64.37 ± 0.21	0.2522 ± 0.0016
$m = 50$	Proposed T1	66.83 ± 0.99	0.2362 ± 0.0104
$m = 50$	Proposed T2	$\mathbf{66.96 \pm 1.16}$	$\mathbf{0.2353 \pm 0.0116}$

TABLE VI
INITIAL REDUCED-DATA MNIST SCREEN ($n = 10{,}000$, $d = 128$, $m = 100$, 25 EPOCHS). THESE ENTRIES USE FIVE INFERENCE DRAWS AND TWO SEEDS, AND ARE KEPT SEPARATE FROM THE TEN-DRAW FINAL COMPARISONS.

Condition	Method	Accuracy (%)	Bounded risk
$m = 100$	ERM	70.25 ± 0.67	0.2007 ± 0.0054
$m = 100$	ℓ_2	69.06 ± 0.47	$\mathbf{0.1939 \pm 0.0047}$
$m = 100$	Proposed T1	$\mathbf{70.82 \pm 0.53}$	0.1971 ± 0.0024
$m = 100$	Proposed T2	70.71 ± 0.45	0.1976 ± 0.0016

The width-64 feature-erasure refinements in Table VII use the same deployment probability 0.5 but different bank sizes. They show the performance attainable with a larger transmitted representation. The proposed variants do not improve on ERM at every bank size. This complements the main finite-channel mechanism, rather than establishing uniform superiority across channels.

B. Data Size and Optimization Budget

Table VIII varies the number of training examples at $m = 100$. Accuracy increases with data size in these runs, but the protocol also increases the number of minibatch updates because it fixes epochs, not updates. The training/validation realizations change with n, and the listed repetitions vary initialization and channel banks rather than resampling independent datasets. Moreover, the observed signed data gap is negative in these realizations. This experiment therefore describes the data-size performance trend; it does not independently verify the theoretical n-rate.

C. Internal Objective Ablations and Selected Cases

The proposed objectives themselves belong to the PAC-Bayesian regularizer family. The data-only penalty is consequently an internal ablation. Both T1 terms depend on the same posterior KL; when that KL dominates the confidence logarithms, the separate penalties are approximately a scaled square-root KL penalty. T2 has a single square-root KL form with adjusted scaling and confidence. An unrestricted retuning of a data-only coefficient can therefore yield near-equivalent optimization. The distinct theoretical contribution is the treatment of finite channel observations and its resulting dependence on n and m.

In the full-data comparison, the validation-selected T1 and data-only configurations attain 76.6307% and 76.6370%, respectively, a practical tie. Allowing test-based hyperparameter selection over the entire trained grid instead gives the data-only control 76.7513%. Completing the missing Fashion-MNIST learning-rate/noise controls reduces the corresponding 100-draw T1 advantage to only 0.0208 percentage point. These findings limit claims of a fundamentally different regularizer.

Selected favorable checkpoints remain useful as illustrative cases. For full-data MNIST at training seed 9001, a new 100-draw stream gives T1 75.9712%, exceeding the strongest of the three reevaluated data-only coefficients at 75.7801%. For the reduced-data $m = 50$ case, seed 1003 gives T2 67.7729% versus 67.4309%. These are explicitly test/seed-selected examples. Their Monte Carlo intervals condition on the fixed checkpoints and test data, and do not establish a population advantage across training seeds or undo selection effects. They are not used in the main mean-performance table.

D. Selected Optimization Settings

Learning rates in the grayscale-image searches are 0.001 and 0.003, with posterior standard-deviation initializations 0.003 and 0.01. The initial L2 search uses coefficients 10^{-5} and 10^{-4} coupled to these two learning rates. The grid is therefore not a full factorial or exhaustive L2 optimization. The initial T1 coefficient pairs are $(\eta_1, \eta_2) = (a, 10a)$ with $a \in \{0.0003, 0.001, 0.003, 0.01\}$, and T2 uses joint coefficients $2a$. Learning rate and initial posterior noise are conditionally assigned, rather than searched as a full Cartesian product. Perfect-channel posterior training considers data coefficients 0.006 and 0.06. Full-data refinement adds T1 pairs $(0.001, 0.01)$ and $(0.003, 0.03)$, and T2 coefficients 0.0015 and 0.004, at learning rate 0.003 and initial posterior standard deviation 0.01. The exact selected values are listed in Table IX. All weights and biases are included in the explicit squared-weight L2 penalty. The complete conditional search configurations are retained in the accompanying reproducibility manifests.

E. CIFAR-10 Screening and Optimization Settings

The preliminary CIFAR-10 screen uses 10,000 training examples, 5,000 validation examples, 30 epochs, and two initialization/channel-bank seeds. Table X retains both seeds of every selected configuration and uses ten fresh deployment

TABLE VII

WIDTH-64 IID TWO-MOONS FEATURE-ERASURE REFINEMENT ($n = 800$, ERASURE PROBABILITY 0.5, 150 EPOCHS). ENTRIES USE TWO SEEDS AND TEN INFERENCE DRAWS. A MATCHED L2 RUN IS UNAVAILABLE FOR THIS COHORT AND IS NOT SUBSTITUTED FROM ANOTHER EXPERIMENT.

Condition	Method	Accuracy (%)	Bounded risk
$m = 5$	ERM	$\mathbf{85.40 \pm 0.11}$	$\mathbf{0.0719 \pm 0.0008}$
$m = 5$	Proposed T1	85.37 ± 0.14	0.0720 ± 0.0008
$m = 5$	Proposed T2	85.37 ± 0.14	0.0720 ± 0.0008
$m = 20$	ERM	$\mathbf{87.72 \pm 0.44}$	$\mathbf{0.0623 \pm 0.0024}$
$m = 20$	Proposed T1	86.18 ± 1.48	0.0684 ± 0.0067
$m = 20$	Proposed T2	86.18 ± 1.48	0.0684 ± 0.0067
$m = 50$	ERM	88.78 ± 0.02	0.0576 ± 0.0005
$m = 50$	Proposed T1	88.80 ± 0.01	0.0575 ± 0.0006
$m = 50$	Proposed T2	$\mathbf{88.80 \pm 0.01}$	$\mathbf{0.0575 \pm 0.0006}$

TABLE VIII

EXPLORATORY DATA-SIZE SLICE FOR T1 ON IID TWO MOONS ($d = 8$, $m = 100$). ACCURACY IS MEAN ± SAMPLE STANDARD DEVIATION OVER THREE INITIALIZATION/CHANNEL-BANK SEEDS. THE SAME 150 EPOCHS IMPLY DIFFERENT OPTIMIZER-UPDATE COUNTS. NEGATIVE OBSERVED DATA GAPS AND THE CHANGING OPTIMIZATION BUDGET PRECLUDE INTERPRETING THIS SLICE AS A QUANTITATIVE VALIDATION OF THE THEORETICAL n-RATE.

n	Updates	Accuracy (%)	Channel gap	Data gap	Data complexity
128	150	71.58 ± 2.18	-0.000306	-0.000147	0.836164
512	600	79.52 ± 2.60	0.000100	-0.002194	0.414271
2048	2400	83.18 ± 0.60	0.000768	-0.006346	0.206496

TABLE IX

VALIDATION-SELECTED SETTINGS FOR TABLE II. THE ACTIVE COEFFICIENTS ARE THE L2 WEIGHT λ_2, T1 CHANNEL/DATA WEIGHTS (η_1, η_2), T2 JOINT WEIGHT η, OR THE PERFECT-TRAINING DATA WEIGHT η_2. THE INITIAL POSTERIOR STANDARD DEVIATION IS s_0; DASHES DENOTE DETERMINISTIC MODELS OR NO PENALTY. POSTERIOR MEANS USE FAN-IN-SCALED UNIFORM INITIALIZATION WITH ZERO BIASES, SEPARATELY FROM s_0; THE FIXED PRIOR HAS STANDARD DEVIATION ONE. THESE ARE SELECTED CONFIGURATIONS FROM CONDITIONAL GRIDS, NOT AN EXHAUSTIVE OR FULL-FACTORIAL SEARCH.

Dataset	Training method	Learning rate	s_0	Active coefficient(s)
MNIST	ERM (perfect)	0.003	–	–
MNIST	ERM (matched)	0.003	–	–
MNIST	ℓ_2 (matched)	0.001	–	$\lambda_2 = 10^{-5}$
MNIST	Proposed (perfect)	0.003	0.01	$\eta_2 = 0.06$
MNIST	Proposed T1 (matched)	0.003	0.01	$(\eta_1, \eta_2) = (0.003, 0.03)$
MNIST	Proposed T2 (matched)	0.003	0.01	$\eta = 0.0015$
FashionMNIST	ERM (perfect)	0.003	–	–
FashionMNIST	ERM (matched)	0.001	–	–
FashionMNIST	ℓ_2 (matched)	0.001	–	$\lambda_2 = 10^{-5}$
FashionMNIST	Proposed (perfect)	0.003	0.01	$\eta_2 = 0.06$
FashionMNIST	Proposed T1 (matched)	0.001	0.01	$(\eta_1, \eta_2) = (0.003, 0.03)$
FashionMNIST	Proposed T2 (matched)	0.001	0.01	$\eta = 0.006$

draws. T1 and T2 exceed ERM and L2 in both seeds, but ERM has lower bounded log loss. The accuracy and bounded-risk orderings therefore differ in this screen.

The 34 training trials cover learning rates 0.001 and 0.003 for ERM; both learning rates crossed with L2 coefficients 10^{-6} and 10^{-5}; and both learning rates crossed with the two proposed settings $(s_0, a) = (0.001, 0.0003)$ and $(0.003, 0.001)$, where s_0 is the initial posterior standard deviation. T1 uses $(\eta_1, \eta_2) = (a, 10a)$ and T2 uses joint coefficient $2a$. Each perfect-training reference has one screened configuration: learning rate 0.001, L2 coefficient 10^{-6} where applicable, and initial posterior standard deviation 0.001 with data coefficient 0.01 for Proposed Perfect. These references are not exhaustively optimized over perfect-training hyperparameters.

The full-data extension freezes the seven validation-selected settings and uses seeds 5101–5103 with 45,000 training examples and 100 epochs. ERM retains learning rate 0.001;

TABLE X

CIFAR-10 DEPLOYMENT PERFORMANCE UNDER GAUSSIAN FADING AND ADDITIVE NOISE. ENTRIES ARE MEAN ± SAMPLE STANDARD DEVIATION ACROSS INITIALIZATION/CHANNEL-BANK SEEDS CONDITIONAL ON THE FIXED DATA SPLIT, USING TEN FRESH GIBBS/CHANNEL DRAWS PER CHECKPOINT. SCREEN: $n = 10,000$, 30 EPOCHS, 2 PAIRED SEEDS. CONFIGURATIONS USE AGGREGATE VALIDATION ACCURACY; CHECKPOINTS USE VALIDATION ACCURACY. PERFECT REFERS TO THE TRAINING FORWARD CHANNEL ONLY. BOLD MARKS THE BEST MEAN WITHIN EACH METRIC.

Method	Screen	
	Accuracy (%)	Bounded risk
ERM (matched)	39.38 ± 0.06	$\mathbf{0.3654 \pm 0.0027}$
L2 (matched)	40.03 ± 2.19	0.4039 ± 0.0146
Proposed T1 (matched)	$\mathbf{44.29 \pm 0.32}$	0.3750 ± 0.0076
Proposed T2 (matched)	43.86 ± 0.73	0.3741 ± 0.0001
ERM (perfect)	29.41 ± 0.36	0.4638 ± 0.0108
L2 (perfect)	29.46 ± 0.63	0.4652 ± 0.0127
Proposed (perfect)	30.44 ± 0.63	0.4488 ± 0.0041

matched L2, T1 and T2 retain 0.003. Matched L2 selects 10^{-6}; T1 selects $(0.001, 0.01)$ and T2 selects joint coefficient 0.002, both with $s_0 = 0.003$. The two stages share the validation prefix and a seeded data permutation, with the smaller training subset nested inside the larger one. They are separate data-size/optimization-budget stages on the same official test set.

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